

REPUBLIC OF YEMEN
University of Aden
Faculty of Computer & Information Technology
ODD Semester Examinations – Academic Year 2020/ 2021

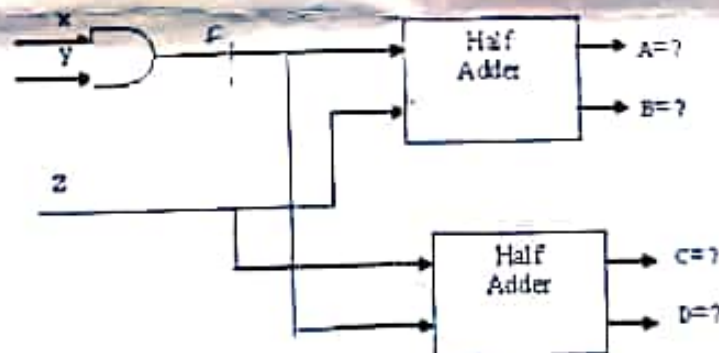
Subject: Logic Design
Class: B2NET

Data:
Time: 2Hours

Note: - Question No. **1** is Compulsory.
- Answer any three questions of the rest

اجباري Q₁

- Simplify the Boolean function $F(a,b,c) = b + bc + ab$ with map method?
- Draw a circuit diagram for the *JK* flip-flop and its truth table.
- Consider the following figure and answer the following question:



Find out the values of A , B , C , and D ?

اختاري Q₂

Design circuit for half-Adder and full-Adder?



Q₅

a. Explain with drawing the differences between synchronous and asynchronous RS flip flop?

b. Combinational logic has 3 inputs (a, b, and c) and 2 output (F1 & F2).

The output F1 equals (1) when the value of inputs > 2

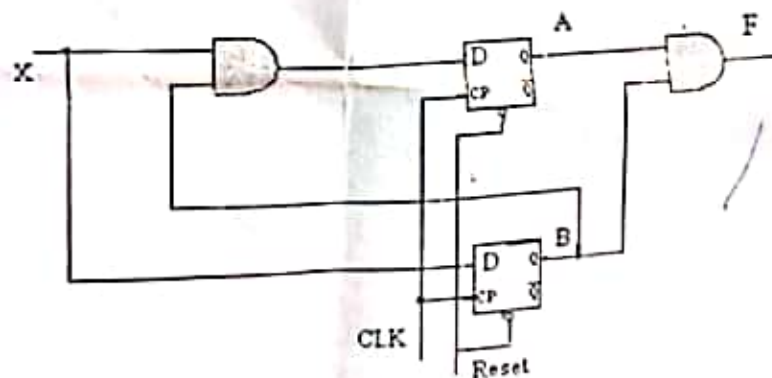
The output F2 equals (0) when the sum of 3-inputs equals 2.

1) Obtain the truth table for F1 and F2.

2) Draw the logic diagram (F1 & F2)

Q₆

Analyze the following clocked sequential circuit. Obtain the state table and state diagram of the sequential circuit.



Good luck

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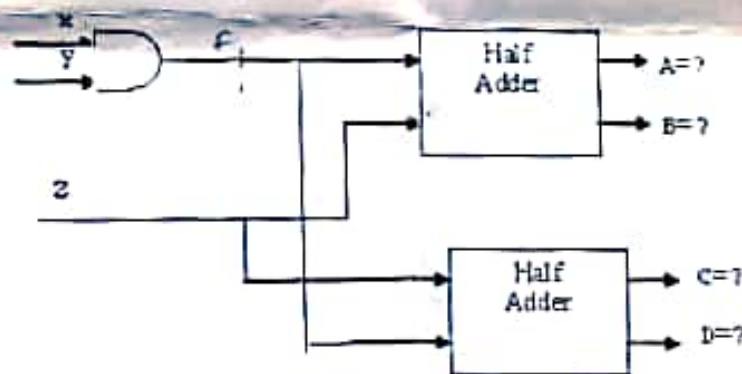
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اختياري Q₂

Design circuit for half-Adder and full-Adder?



Q3

a. Implement the following function with multiplexer

$$F(x,y,z) = \sum (2,3,5,6,7)$$

b. How do you get Dec 4*16 by using Dec 3*8?



x	y	z	
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7

Q4

a. Explain with drawing the differences between synchronous and asynchronous RS flip flop?

b. Combinational logic has 3 inputs (x, y, and z) and 2 output (F1 & F2).

The output F1 equals (0) when the value of inputs > 4

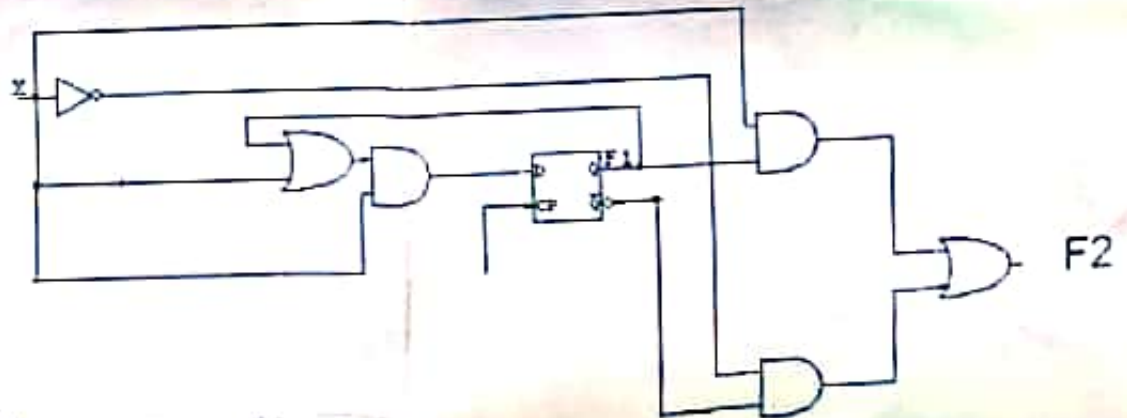
The output F2 equals (0) when the sum of 3-inputs equals 2.

1) Obtain the truth table for F1 and F2.

2) Draw the logic diagram (F1 & F2)

Q5

. Analyze the sequential circuit, and obtain F2?



Q4) v)

x	y	z	F1	F2
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

Good luck

x	y	z	F1	F2
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0